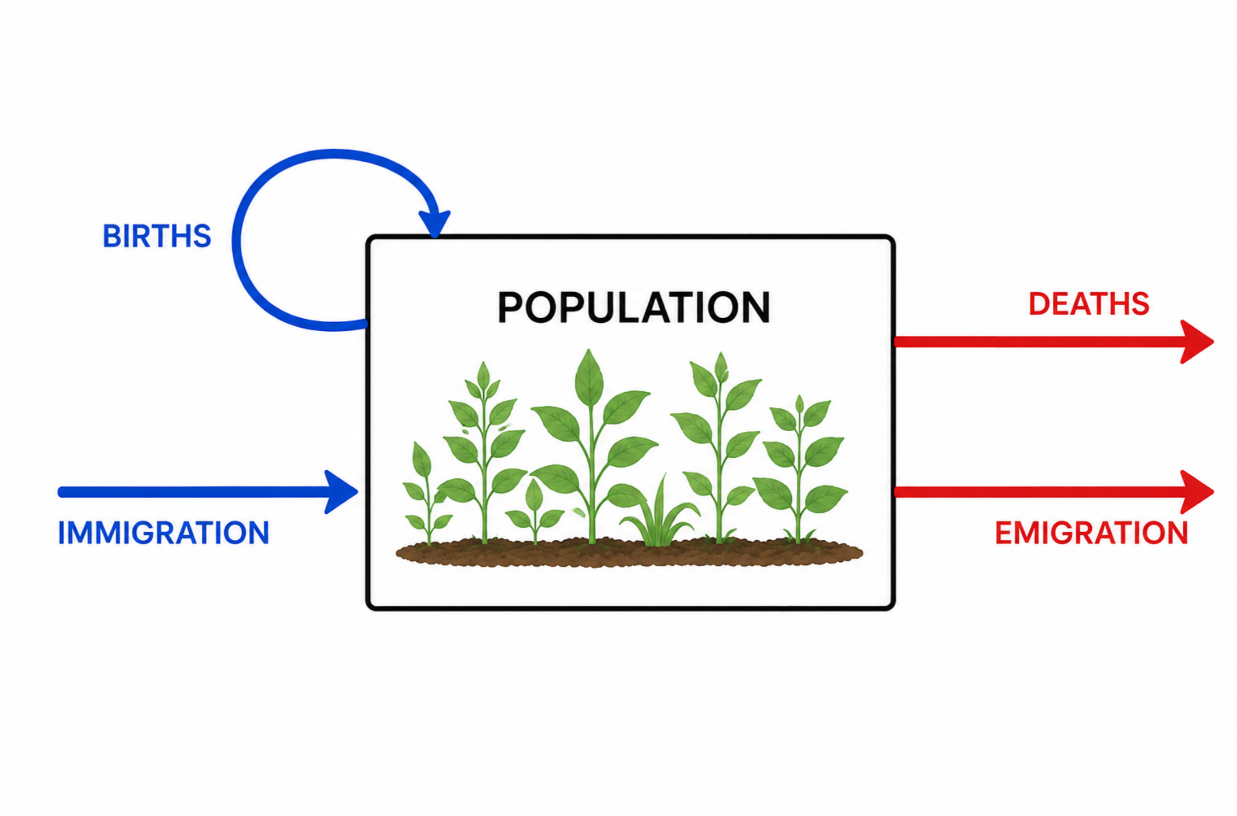
Population growth

Learning objectives

1. Understand basic processes influencing population size
2. Plot population sizes over time wiht different growth rates
3. Model a population with discrete and continuous time

in most basic from, population size is determined by births, deaths, immigration, and emigration



births are a feedback loop because births are coming from the population

we are most interested in population dynamics over time

- understand causes of variation
- predict population sizes across space and time

Terminology and variables in population models

- data: time series *tracking population sizes over time*
- discrete time - divide into time steps (e.g. year), difference equation
- continuous time - differential equation

we'll mostly work with discrete time

relate to Kezia's statistics class- geometric distribution for discrete response variables, exponential distribution for continuous response variables, but very similar

Discrete time

- N = population size
- t = time
- N_t = population at t
- N_{t+1} is size in next year*
- λ = per capita *per individual* growth rate

$$N_{t+1} = \lambda N_t$$

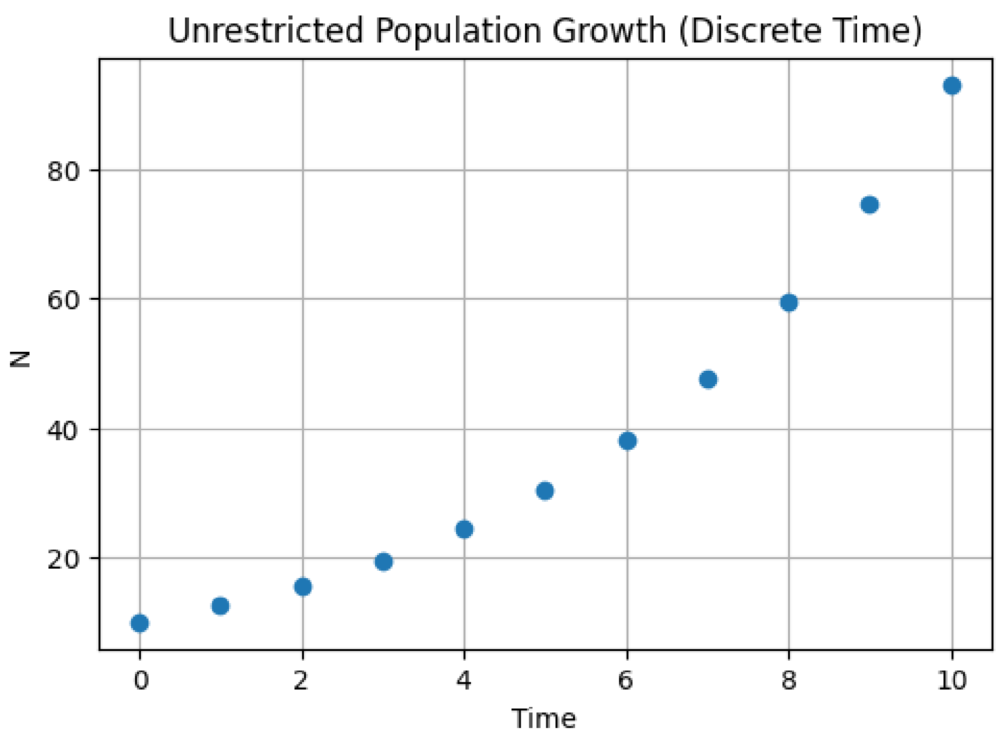
- $\lambda = 1$, population growth is constant
- $\lambda < 1$, population decline
- $\lambda > 1$, population growth

we can rearrange this equation so we get population at any t

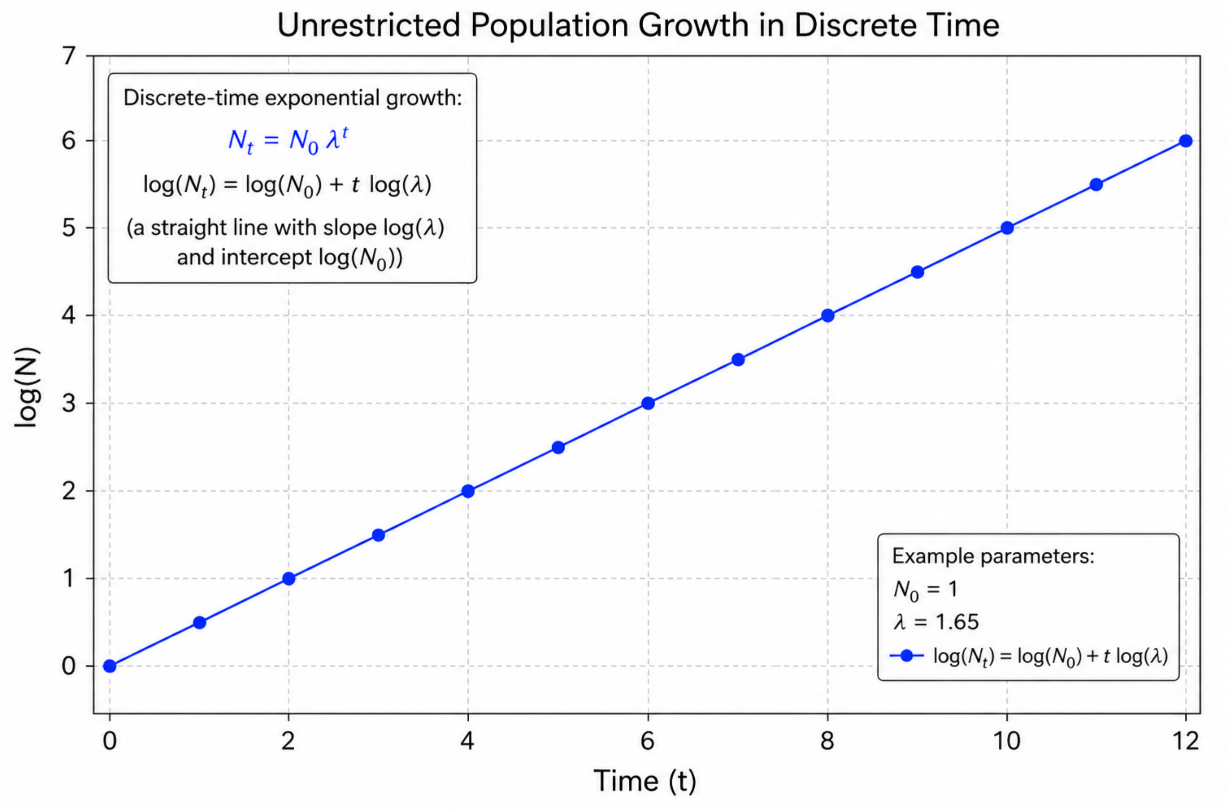
$$N_t = N_0 \lambda^t$$

N_0 is initial population size

with that exponent, looks more like unrestricted growth like what we would expect



if we log the y axis, this relationship becomes linear



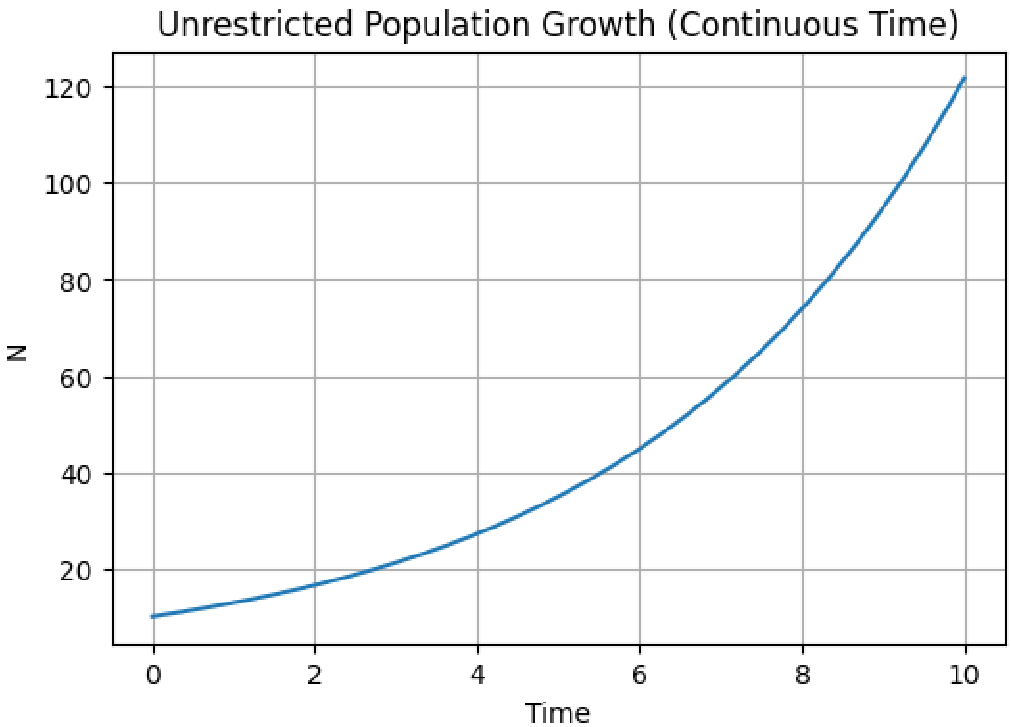
Continuous time with exponential growth

- N
- t
- dN/dt = rate of change in population size
- r = intrinsic growth rate (continuous, exponential)

$$\frac{dN}{dt} = rN(t)$$

- r = 0, change in population size is 0
- -r, population decline
- +r, population growth
- $\log(\lambda)$ = r *you may see r with discrete time models, so this is the relationship*

$$N(t) = N(0)e^{rt}$$



in real populations, births, deaths, immigration, emigration aren't constant

so population sizes over time aren't nice and continuous

Demographic stochasticity

- variation due to chance events affecting individuals
- birth and death rates are an average
- individual births and deaths can't be fractions *when applied to a population of 5, death rate of 0.5 is average, not 2.5 individuals survive*
- per capita growth rate comes from a distribution *e.g. lambda is uniform distribution between 0.5 and 1.5*
- increases extinction risk in small populations *because if by chance reproductive individuals have few offspring for a few years, population could go extinct*
- can ignore in large populations

Geometric vs. arithmetic growth

given that lambda fluctuates year to year, what if you wanted to get the average population growth rate?

lambda is per capita population growth rate, so we need average lambda

exponential-looking relationship makes it confusing when we're trying to get population growth rate

- average population growth rate = mean(log(λ)) = geometric mean
- log(mean(λ)) = arithmetic mean *can overestimate population growth*
- consideration for time but not space *ask for guesses about why this is the case?*
- compounding: annual growth multiples the population from previous year *multiplicative growth not additive growth*